AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System

System Architecture, Base Paper Integration, and Implementation Specification

1. Executive Summary & Project Objective

In e-commerce, consumers make purchasing decisions based entirely on digital product representations (titles, specifications, bullet points, and catalog photos). However, discrepancies between advertised claims and delivered products (material composition, sizing errors, visual color shifts, and operational failures) remain a widespread cause of customer dissatisfaction and high product return rates.

PROJECT OBJECTIVE: This project engineers an automated, multimodal consistency assessment system. The platform ingests seller listings and cross-examines them against empirical customer evidence (written reviews and customer-uploaded unboxing photographs). The system executes an objective Quality Assurance (QA) audit, quantifying support and contradiction for each claim and computing a unified Product Representation Index (PRI).

2. Base Paper Integration: PRAISE (ACL 2025)

BASE PAPER CITATION:
Title: PRAISE: Enhancing Product Descriptions with LLM-Driven Structured Insights
Authors: Adnan Qidwai, Srija Mukhopadhyay (IIIT Hyderabad), Prerana Khatiwada (Univ. of Delaware), Dan Roth (UPenn), Vivek Gupta (Arizona State University)
Venue: ACL 2025 (Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics), System Demonstrations, pp. 644-652. (Prestige: Core A*, #1 NLP Venue Globally)

What Our System Adopts from PRAISE:

1. Factual Attribute Extraction: Extracting verifiable attribute-value pairs from reviews while filtering subjective noise.
2. Cross-Source Alignment: Directly comparing extracted review facts against official seller descriptions.
3. Four-Class Categorization Schema: Matching, Contradictory, Missing, and Partially-Matching.
4. Benchmark Dataset Foundation: Multi-category Amazon Review corpora (Apparel, Footwear, Electronics, Appliances).

3. Architectural Extensions Over the Base Paper

Dimension	Base Paper: PRAISE (ACL 2025)	Our Proposed Implementation
Modality Scope	Strictly Text-Only (Documented limitation in Sec 8)	Multimodal (Text reviews + Customer unboxing photos)
Visual Verification	None	CIE76 Delta E (LAB color space) + CLIP visual similarity
Execution Efficiency	~ 2R + 1 brute-force LLM calls (200+ calls for 100 reviews)	Sentence-BERT Top-K semantic retrieval (cuts latency by 80%)
Output Formulation	Categorical tables only	Product Representation Index (PRI: 0-100%) + 4-tier taxonomy
System Interface	Basic research demonstration	Interactive Streamlit Dashboard with color swatches & audits

4. End-to-End Implementation Pipeline

Module 1: Claim Deconstruction Engine

Deconstructs seller descriptions into atomic, testable propositions (Tp) with category tags:
Example: '100% Genuine Italian Leather Shoes with non-slip rubber sole. Fits true to size.'
-> T1: 'Upper material is 100% Italian leather' [MATERIAL] | T2: 'Outsole is slip-resistant rubber' [FUNCTION] | T3: 'Fits true to size' [SIZING].

Module 2: Semantic Evidence Retrieval Layer (Sentence-BERT)

Encodes all customer review sentences using sentence-transformers/all-MiniLM-L6-v2 into 384-dimensional vectors. Computes cosine similarity against claim Tp and retrieves Top-K (5-10) candidate sentences, filtering non-relevant commentary.

Module 3: Text Natural Language Inference (NLI Engine)

Pairwise cross-attention between review sentence (Premise) and claim (Hypothesis) via cross-encoder/nli-roberta-base. Outputs probability distributions: P(Support), P(Contradict), P(Neutral). Aggregates into Global Support S(Tp) and Contradiction C(Tp).

Module 4: Multimodal Vision Engine (Modular Extension)

When customer photos are uploaded, the vision engine computes:
- Perceptual Color Distance (CIE76 Delta E): Converts dominant colors to CIE LAB space (human perceptual space). Flags Delta E > 25.
- Structural Visual Similarity (CLIP): Generates OpenAI CLIP ViT-B/32 image embeddings to evaluate silhouette and pattern consistency.
Modularity: If no customer photos exist, vision bypasses gracefully (beta = 0); the text engine operates at 100% independently.

Module 5: Reliability Synthesis & Product Representation Index (PRI)

Mathematically aggregates text scores and visual discrepancy into a unified quantitative metric:
PRI = 100 * [alpha * S_text + (1 - alpha)*(1 - C_text) - beta * D_visual]. Assigns 4-tier status: Consistent, Contradictory, Mixed, Insufficient.

5. Mathematical Formulations & Metrics

1. Sentence Credibility Weight: w_i = 1.0 + (0.50 * is_verified) + (0.05 * min(helpful_votes, 10))
2. Global Support Score: S(Tp) = [sum(w_i * P(Support_i))] / sum(w_i)
3. Global Contradiction Score: C(Tp) = [sum(w_i * P(Contradict_i))] / sum(w_i)
4. CIE76 LAB Color Difference: Delta E = sqrt((L_catalog - L_user)^2 + (a_catalog - a_user)^2 + (b_catalog - b_user)^2)
5. Visual Discrepancy Score: D_visual = 0.5 * min(1.0, Delta E / 50) + 0.5 * (1.0 - Sim_CLIP)
6. Product Representation Index: PRI = 100 * [alpha * S_mean + (1 - alpha)*(1 - C_mean) - beta * D_visual] (alpha=0.6, beta=0.15)

6. Edge-Case Handling & Robustness Matrix

Operational Condition	System Detection & Decision Rule	System Output & Action
Cold Start (N = 0 Reviews)	Zero customer feedback corpus available.	Insufficient Evidence: Triggers Intrinsic Claim Audit flagging unprovable superla
Low Sample Size (N < 3)	Reviews exist but lack statistical mass.	Preliminary Assessment: Displays preliminary scores with an explicit low-sampl
Bimodal Sizing Feedback	S(Sizing) >= 0.40 and C(Sizing) >= 0.40.	Mixed Evidence: Bimodal distribution displayed (e.g., '48% report true-to-size; 5
High Rating, False Spec	5-star rating, but C(Material) >= 0.85.	Contradictory Claim: Flags specific specification contradiction regardless of pos
Ambient Lighting Variance	Delta E = 16 (Moderate), but CLIP Sim = 0.91.	Consistent with Lighting Note: System identifies illumination artifact, avoiding fa
Aesthetic Match, Dead Unit	Photos match (Delta E < 8), but C(Battery) = 0.92.	Functional Contradiction: Confirms aesthetic alignment while isolating operation
Missing Customer Photos	Reviews exist, zero user images uploaded.	Modular Bypass: beta set to 0.0; text consistency engine runs at 100% capacity

7. Software Architecture & Implementation Modules

The project is structured into clean, modular components:
- src/data_loader.py: Ingests and normalizes Amazon Reviews JSON metadata (catalog + review records).
- src/claim_extractor.py: Deconstructs seller paragraphs into atomic proposition objects (Tp).
- src/retriever.py: S-BERT dense vector index and Top-K cosine semantic filtering.
- src/nli_engine.py: RoBERTa cross-encoder inference for Support, Contradict, and Neutral probabilities.
- src/visual_comparator.py: OpenCV LAB color conversion, K-Means clustering, Delta E, and CLIP embeddings.
- src/scoring_engine.py: Mathematical calculation of credibility weights w_i, S, C, PRI, and 4-tier taxonomy.
- app.py: Streamlit dashboard rendering interactive consistency dials, claim breakdowns, and evidence quotes.

8. Real-World Stakeholder Utility

1. Consumers (Shoppers): Immediate 3-second claim reliability breakdown prior to checkout, preventing bad purchases.
2. E-Commerce Platforms (Amazon/Flipkart): Reduces reverse logistics expenses in fashion/footwear where return rates reach ~30% due to 'item not as described' and size/color mismatches; ensures compliance with CCPA/FTC guidelines.
3. Merchants & Sellers: Generates actionable diagnostic reports (e.g., '40 customers report shoulder tightness; update size chart'), helping merchants refine listings and curb return volumes.